

Analysis for Anions

You will be testing the unknown substance for the presence of a number of common anions. Explained below are the tests along with some detailed notes on special experimental techniques where necessary.

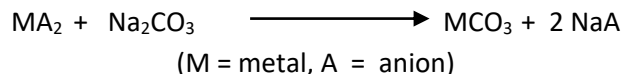
In general, anions can be classified in to two different categories:

- a) Anions that evolve volatile products upon treatments with acids – these are further sub divided into two groups :
 - i. Anions that evolve volatile gases with dil HCl or dil H₂SO₄ (CO₃²⁻, SO₃²⁻, S₂O₃²⁻, S²⁻, NO₂⁻, ClO⁻, CN⁻, CNO⁻)
 - ii. Anions that evolve gases with conc.H₂SO₄ (all of the above, and F⁻, Cl⁻, Br⁻, I⁻, NO₃⁻, ClO₄⁻, MnO₄⁻, BrO₃⁻, ClO₃⁻, Fe(CN)₆⁴⁻, Fe(CN)₆³⁻, SCN⁻, C₂O₄²⁻)
- b) Anions that can be identified on the basis of their reactions in solutions, which are also sub-divided into two groups:
 - i. Anions that undergo precipitation reactions in solution (SO₄²⁻, S²⁻, PO₄²⁻, BO₃³⁻, B₄O₇²⁻, BO₂⁻, H₂PO₂⁻, AsO₃³⁻, AsO₄³⁻, CrO₄²⁻, Cr₂O₇²⁻, SiO₃²⁻, SiF₆²⁻, C₆H₄(OH)CO₂⁻, C₄H₄O₄²⁻) and
 - ii. Anions that undergo redox (oxidation – reduction) reactions in solution (MnO₄²⁻, MnO₄⁻, CrO₄²⁻, Cr₂O₇²⁻)

Note: For all the known sodium salts, prepare 1 mL solution by dissolving 0.05 g – 0.1 g of the salt and carryout all the analysis given below for 5 drops of the solution by adding 1 – 2 drops of the reagent mentioned. If the salt is known, Na₂CO₃ extraction should be prepared before the analysis for anions.

Preparation of a Na₂CO₃ extract /Soda extract

Mix the anhydrous Na₂CO₃ extract 3-4 times .Dissolve 0.6 g of anhydrous Na₂CO₃ in 10 mL of distilled water. Boil about 0.2 g of the mixture with this solution for 10 min in a boiling tube (10 mL), centrifuge/filter and decant off the clear solution. Use the extract to carry out the tests for anions.



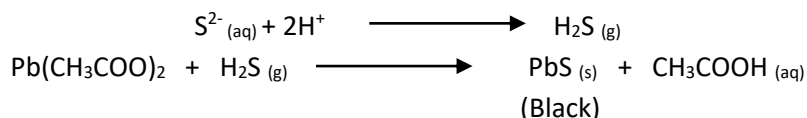
Then, the solution containing the anions, but free from metals, may be filtered off and used for subsequent tests for anions.

Test for Anions

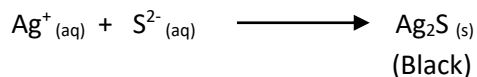
1. Test for sulfides (S^{2-})

$Na_2S \cdot 9H_2O$ is used and the following tests are carried out using an aqueous solution.

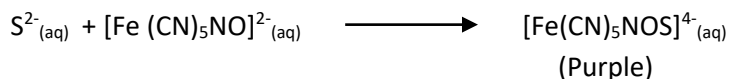
- a. Dill H_2SO_4 (1-2 Drops) is added to the test solution and the evolved gas is tested with a filter paper dipped in a solution of lead acetate. Here, H_2S having an unpleasant odor (commonly called the 'rotten egg' smell) results, which then reacts with lead acetate giving black lead sulphides.



- b. Silver nitrate (1-2 drops) is added to the solution (5 drops), and the solubility of the resulting precipitate is tested in cold and hot dill. HNO_3 . A black precipitate of silver sulfide forms which is insoluble in cold dill HNO_3 , but dissolves upon warming.



- c. A few drops of sodium hydroxide solution and then some freshly prepared sodium nitroprusside solution are added to the test solution. A purple coloration is observed upon the addition of reagents.

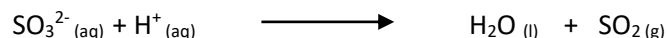


Note: This can be seen only in the solutions of alkalis. No reaction occurs with solutions of H_2S or with free gas

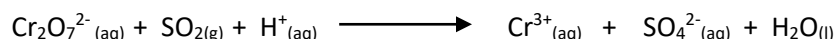
- d. Barium chloride (1-2 drops) is added to the test solution (5 drops). Unlike in the case of sulphates, no precipitate is observed. This is, therefore, a good test to differentiate between sulphates and sulphides. However, sulphides, especially if wet, undergo air oxidation to give sulphites and eventually sulphates, and hence a small amount of an acid insoluble precipitate may be obtained when this test is performed.

2. Tests for sulphites (SO₃²⁻)

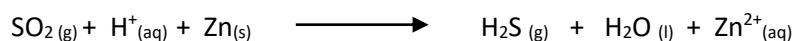
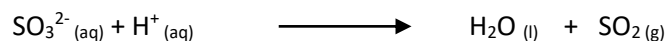
- a. Na₂SO₃·7H₂O is used and the following tests are carried out for sulphites using an aqueous solution. Dill. H₂SO₄ (1-2 drops) is added to 5 drops of the solution of the sulphite and, warmed. The gas-evolved is tested using a moistened litmus paper and a piece of filter paper is dipped in acidified dichromate solution.



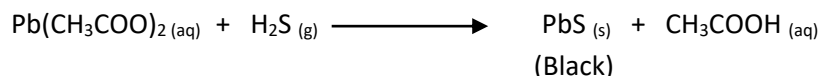
Sulphur dioxide is acidic and therefore a moistened piece of litmus paper would turn red upon contact with SO₂. It also reduces dichromate (orange) to Cr³⁺ (green).



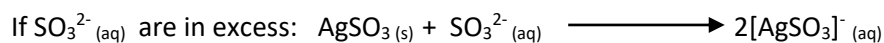
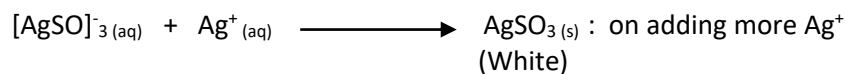
- b. A few granules of zinc and dill H₂SO₄ is added to the test solution, and the gas evolved is tested with a piece of filter paper dipped in a solution of lead acetate. The SO₂ produced as indicated above get reduced by Zn to H₂S.



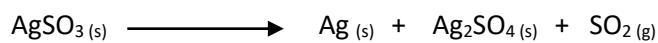
The H₂S turns lead acetate paper black.



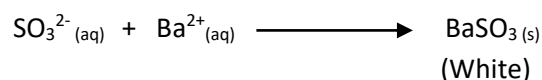
- c. Silver nitrate (1-2 drops) is added to the test solution (5 drops) and boiled.



The white precipitate initially formed turns grey or black upon heating due to the formation of metallic silver.



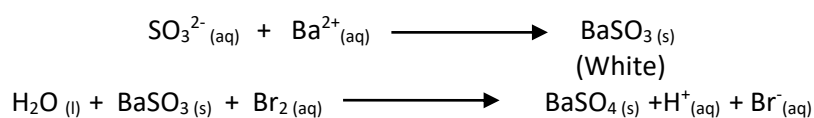
- d. Barium chloride (1-2 drops) is added to the test solution (5 drops), and the solubility of the resulting precipitate is tested in dill. HNO_3



The white precipitate of BaSO_3 is soluble in dill. HNO_3 .

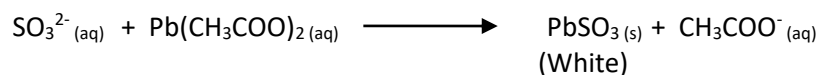
On standing, $\text{BaSO}_3(\text{s})$ is slowly oxidized to $\text{BaSO}_4(\text{s})$ and is then insoluble in mineral acids.

- e. A little bromine water, barium chloride solution and dill. HNO_3 is added to the test solution. The white precipitate of BaSO_3 is oxidized by Br_2 to BaSO_4 which is also a white precipitate but insoluble in dill acids. Since bromine undergoes reduction to form Br^- the solution gets decolorized.

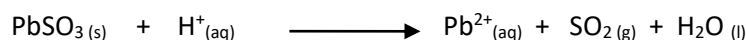


Note: air, Br_2 , conc. HNO_3 and H_2O_2 facilitates the oxidation of $\text{BaSO}_3(\text{s})$ to $\text{BaSO}_4(\text{s})$

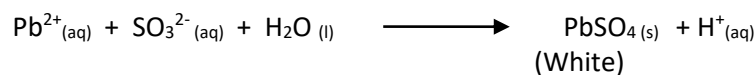
- f. A solution of lead acetate (1-2 drops) is added to the test solution, and the solubility of the resulting precipitate is tested in dill. HNO_3 .



This white precipitate is soluble in dill. HNO_3



Upon boiling this solution, an insoluble white precipitate is obtained.

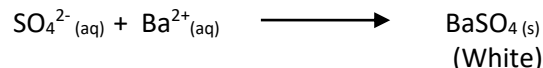


This reaction can be distinguish sulphites and thiosulphates: Thiosulphate produces a black precipitate.

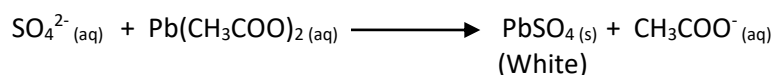
3. Test for Sulphates (SO_4^{2-})

$\text{Na}_2\text{SO}_4 \cdot 10 \text{H}_2\text{O}$ is used and the following tests are carried out for sulphates using an aqueous solution.

- a. Barium chloride (1-2 drops) is added to the test solution (5 drops), and the solubility of the resulting precipitate is tested in dil. HNO_3



- b. A solution of lead acetate (1-2 drops) is added to the test solution (5 drops), and the solubility of the resulting precipitate is tested in dill. HNO_3

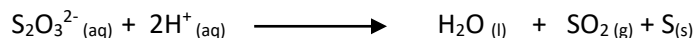


This white precipitate of lead sulphate is insoluble in dill. HNO_3

4. Test for thiosulphates ($\text{S}_2\text{O}_3^{2-}$)

$\text{Na}_2\text{S}_2\text{O}_3 \cdot 5 \text{H}_2\text{O}$ is used and the following tests are carried out using an aqueous solution.

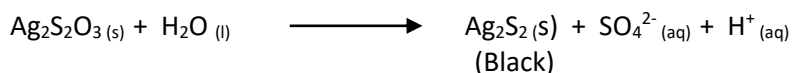
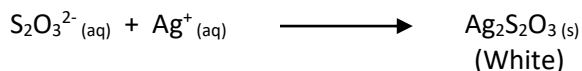
- a. Dill H_2SO_4 (1-2 Drops) is added to the test solution (5 drops), warmed and the gas evolved is tested for SO_2 as mentioned above.



- b. A solution of silver nitrate (1-2 drops) is added to the test solution (5 drops). The white precipitate of $\text{Ag}_2\text{S}_2\text{O}_3$ dissolves rather easily if an excess of thiosulphate is present.

At first, no precipitate occurs due to the formation of $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$

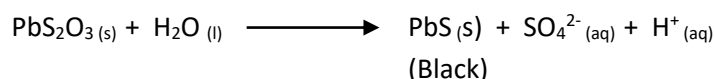
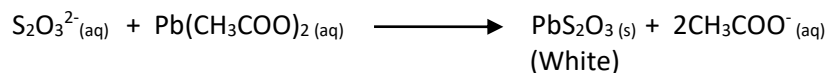
An unstable white precipitate forms initially which decomposes to give a black precipitate of Ag_2S .



- c. Iodine solution (1-2 drops) is added to the test solution (5 drops). The colour of iodine disappears since it oxidizes thiosulphate to tetrathionate.



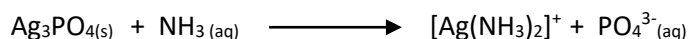
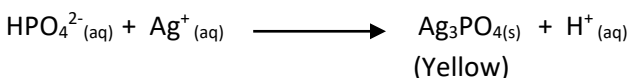
- d. Lead acetate (1-2 drops) is added to the test solution (5 drops). A white precipitate results which is soluble in excess of lead acetate, and on boiling, a black precipitate of lead sulphide is formed.



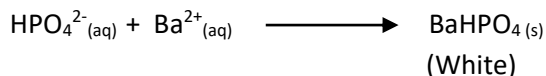
5. Tests for phosphates (PO_4^{3-})

$\text{Na}_2\text{HPO}_4 \cdot 12 \text{H}_2\text{O}$ is used and the following tests are carried out using an aqueous solution.

- a. Silver nitrate (1-2 drops) is added to a neutral test solution (5 drops). A yellow precipitate of Ag_3PO_4 which is soluble in both dill. HNO_3 and in ammonia is formed.



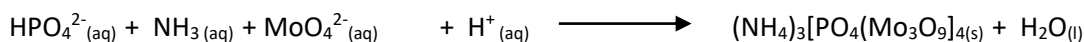
- b. Barium chloride (1-2 drops) is added to a neutral test solution (5 drops). A white precipitate of BaHPO_4 which is soluble in dill. HCl , dill HNO_3 and acetic acid is obtained.



In the presence of dill. NH_3 the less soluble tertiary phosphate is precipitated.



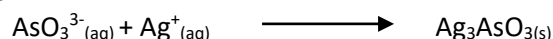
- c. Conc. HNO_3 and ammonium molybdate 2-3 mL is added to a 0.5 mL of the test solution and warmed. A yellow precipitate of ammonium phosphomolybdate $(\text{NH}_4)_3[\text{PO}_4(\text{Mo}_3\text{O}_9)_4]$ is obtained. If the solution is dilute, only a yellow coloration of solution will be seen.



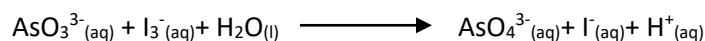
Note: reducing agents such as sulphides, sulphites, etc. seriously affect the reaction and should be destroyed before carrying out this test.

6. Tests for Arsenites (AsO_3^{3-})

- a. Silver nitrate (1-2 drops) is added to a neutral test solution (5 drops). A yellow precipitate of Ag_3AsO_3 is obtained.

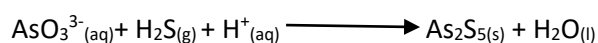
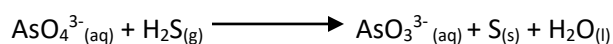


- b. Add small amount of potassium triiodide to test solution. Then sodium hydrogen carbonate is added to complete the reaction.

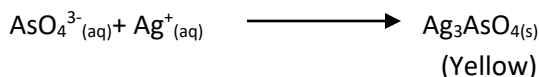


7. Tests for Arsanates (AsO_4^{3-})

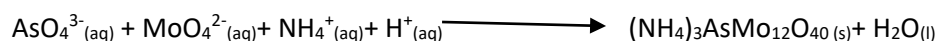
- a. Bubble $\text{H}_2\text{S}_{(g)}$ to the test sample. No immediate precipitate is obtained in the presence of dil HCl acid. When the passage of gas is continued, mixture of $\text{As}_2\text{S}_3_{(s)}$ and $\text{S}_{(s)}$ slowly precipitated.



- b. Add little amount of Silver nitrate to the test sample. The precipitate is soluble in acids and in ammonia solution and insoluble in acetic acid.



- c. Excess amount of Ammonium molybdate solution and nitric acid are added to the test sample. Then boil the solution. The precipitate is insoluble in nitric acid and soluble in ammonia solution and in solutions of caustic alkalis.

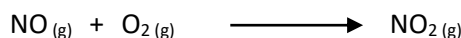
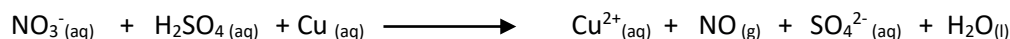


8. Test for Nitrates (NO_3^-)

- a. Concentrated sulphuric acid is added and heating the solid nitrate with the reagent. Reddish brown vapour of nitrogen dioxide, and pungent acid vapour of nitric acid which fumes in the air, are formed. Diluted sulphuric acid has no effect.



- b. Concentrated sulphuric acid and bright copper turnings on heating these with the solid nitrate, reddish – brown fumes of nitrogen dioxide are evolved, and the solution acquires blue colour owing to the formation of copper (II) ions. A solution of the nitrate may also be used: the sulphuric acid is then added very cautiously.



- c. Brown ring test : This test is carried out in either of two ways

- I. Add 3 ml freshly prepared saturated solution of iron (II) sulphate to 2 ml nitrate solution, and pour 3-5 ml Concentrated sulphuric acid slowly down the side of the test tube so that the acid forms a layer beneath the mixture. A brown ring will form where the liquids meet.

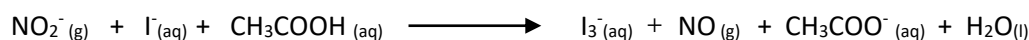
Or

- II Add 4 ml Concentrated sulphuric acid slowly to 2 ml nitrate solution, mix the liquids thoroughly and cool the mixture under a stream of cold water from the tap. Pour a saturated solution of iron (II) sulphate down the side of the tube so that it forms a layer on top of liquid. A brown ring will form at the zone of contact of the two liquids.

The brown ring is due to the formation of the $[\text{Fe}(\text{NO})]^{2+}$. On shaking and warming ions must be absent.

- d. The addition of a nitrate solution to a solution of potassium iodide, followed by acidification with acetic acid or with dilute sulphuric acid, results in the liberation of iodine, which may be identified by the blue colour produced with starch paste.

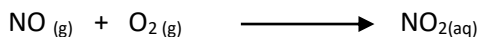
A similar result is obtained by dipping potassium iodide-starch paper moistened with a little dilute acid into solution.



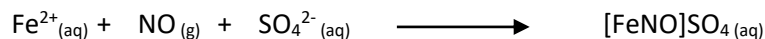
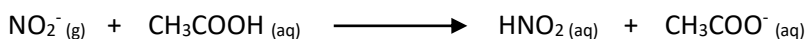
9. Test for Nitrites (NO_2^-)

- a. Cautious addition of the acid to a solid nitrite in the cold yields a transient, pale – liquid and the evolution of brown fumes of nitrogen dioxide, the later being largely produced by combination of nitric oxide with the oxygen of the air. Similar results are obtained with the aqueous solution.



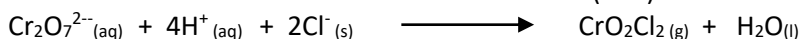
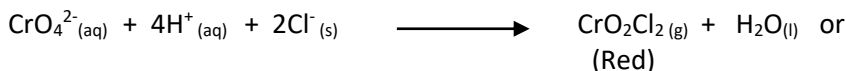


- b. When the nitrite solution is added carefully to a saturated solution of iron (II) sulphate acidified with dilute acetic acid or with dilute sulphuric acid, a brown ring, due to the compound $[\text{FeNO}]\text{SO}_4$, is formed at the junction of the two liquids. If the addition has not been made cautiously, a brown coloration results. This reaction is similar to the brown ring test for nitrates, for which a stronger acid must be employed.

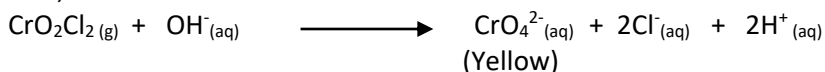


Iodide, bromides, colored ion and anions that give colored compounds with iron (II)

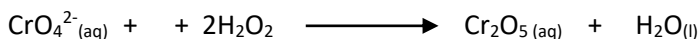
- c. To three parts of the solid compound, one part of sodium chloride and a few drops of conc. H_2SO_4 are added and warmed gently. When the red vapour of chromyl chloride formed is passed in to a solution of sodium hydroxide (colorless), sodium chromate (Yellow) is formed.



And then,



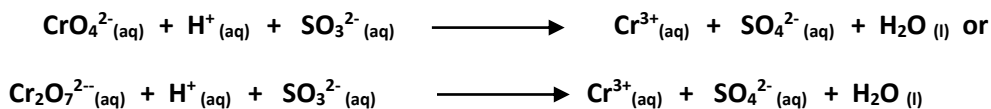
If this solution is tested with a H_2O_2 in amyl alcohol, a blue colouration is seen due to the formation of Cr_2O_5 :



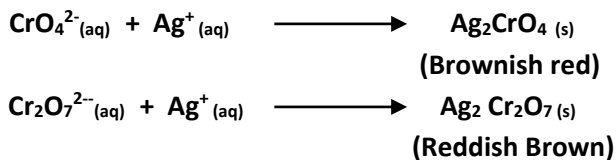
10. Test for chromates and dichromates ($\text{CrO}_4^{2-}/\text{Cr}_2\text{O}_7^{2-}$)

Use either $\text{K}_2\text{Cr}_2\text{O}_4$ or $\text{K}_2\text{Cr}_2\text{O}_7$ and the following tests are carried out using aqueous solutions.

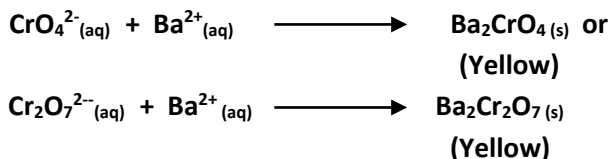
- a. Dilute H_2SO_4 and dilute H_2SO_3 (in excess) are added to a neutral test solution (5 drops). A green colored solution of Cr^{3+} is obtained.



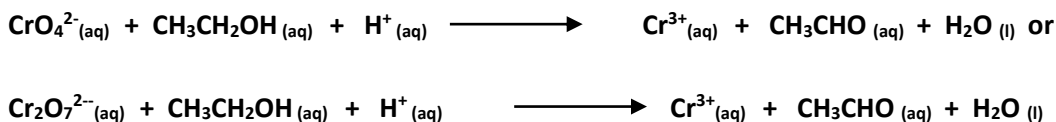
- b. Silver nitrate (1-2 drops) is added to the test solution (5 drops). A brownish red precipitate of silver chromate or a reddish brown precipitate of silver dichromate is obtained, which is soluble in dill. HNO_3



- c. Barium chloride (1-2 drops) is added to the test solution (5 drops). A yellow precipitate of barium chromate (or barium dichromate) is obtained, which is soluble in dill. HNO_3 but insoluble in dill. $\text{CH}_3\text{COOH}(\text{aq})$



- d. Dill. H_2SO_4 (1-2 drops) and ethyl alcohol (1-2 drops) are added to the solution and warmed. A green solution of a Cr^{3+} salt and a characteristic smell of typical aldehydes should result.



The mixture the brown colour disappears, nitrogen oxide is evolved, and a yellow solution of iron remains. The test is unreliable in the presence of bromide, iodide, chlorate, and chromate.



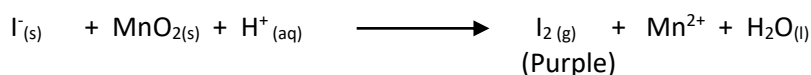
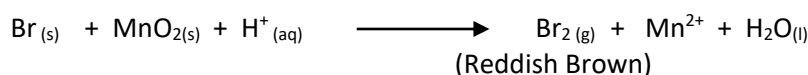
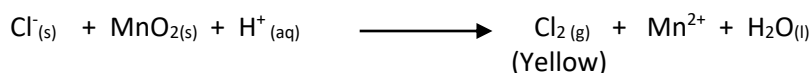
When a solution of a nitrate is boiled with zinc dust or gently warmed with aluminum powder and sodium hydroxide solution. Excellent results are obtained by the use of Devarda's alloy. In alkaline medium

Ammonia is evolved by its odor: by its action upon red litmus paper and upon mercury(I) nitrate paper.

11. Tests for halides (Cl⁻, Br⁻, I⁻)

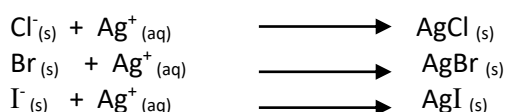
NaCl, NaBr are used and the following tests are carried out using aqueous solutions.

- a. To the solid compound (0.05 g) approximately 0.05 g of manganese dioxide and conc. H₂SO₄ (few drops) are added and warmed, and the gases evolved are tested with moistened litmus paper. Free halogen forms in this reaction. All three, gases bleach litmus paper. Chlorine gas is yellowish green in colour, bromine gas is reddish brown and iodine is purple. Gaseous chlorine and bromine, in addition, have something odors.



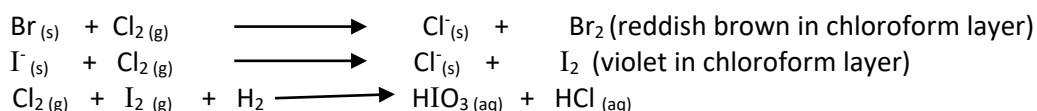
- b. To one part of the solid compound, three parts of K₂Cr₂O₇ and a few drops of conc. H₂SO₄ are added and warmed, and the gases evolved are passed into a solution of sodium hydroxide. This solution is tested by, adding a solution of barium chloride, adding amyl alcohol and hydrogen peroxide, and adding chloride water and chloroform. Bromides and iodides give rise to the free halogens, which yield pale yellow solutions (Bromides) or colorless solutions with sodium hydroxide.

- c. Silver nitrate is added to the test solution. Chlorides give a white precipitate, bromides a pale yellow precipitate and iodides a curdy yellow precipitate of AgI



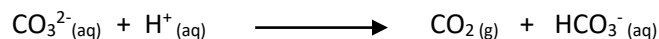
AgCl is insoluble in dill. HNO₃ but soluble in dil. Ammonia. AgBr is insoluble in dill. HNO₃, Sparingly soluble in dilute ammonia, but readily soluble in conc.ammonia. AgI is insoluble in dill. HNO₃ and sparingly soluble in conc.ammonia.

- d. Chloroform/Carbon Tetrachloride is added to the test solution and chlorine water is then added dropwise, followed by an excess. The test tube is shaken vigorously. Chlorides do not show any reaction. In the case of bromide, Br₂ is generated which colors the organic layer violet and with excess of chlorine water the iodine in the organic layer is converted to colorless iodic acid.

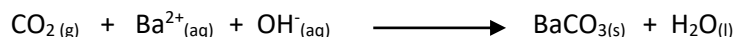
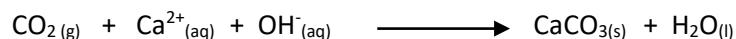


12. Test for Carbonates (CO_3^{2-})

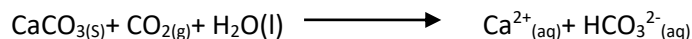
- a. Dilute hydrochloric acid decomposition with effervescence, due to the evolution of carbon dioxide.



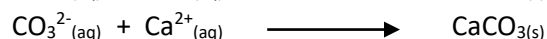
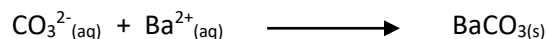
The gas can be identified by its property of rendering lime water (or baryta water)



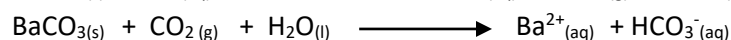
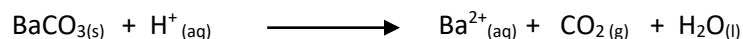
The gas which is evolved (warming may be necessary) is passed into lime water or baryta water contained in the test tube, the production of turbidity indicates the presence of a carbonate. It must be remembered that, with prolonged passage of carbon dioxide, the turbidity slowly disappears as a result of the formation of a soluble hydrogen carbonate.



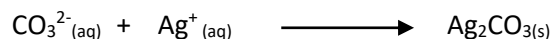
- b. Barium chloride (or calcium chloride) solution is added and white precipitate of barium (or calcium) carbonate (soda water):



Only normal carbonates react: hydrogen carbonates do not. The precipitate is soluble in mineral acids and carbonic acid (soda water):



- c. Silver nitrate solution is added, white precipitate silver carbonate formed.



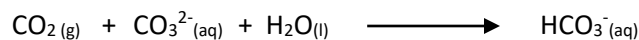
The precipitate is soluble in nitric acid and in ammonia. The precipitate becomes yellow or brown upon addition of excess reagent owing to the formation of silver oxide: this happens if the mixture is boiled:



The solution obtained after dissolving in ammonia should be discarded quickly to avoid a serious explosion.

- d. Sodium carbonate – phenolphthalein test

This test depends upon the fact that phenolphthalein is turned pink by soluble carbonate and colorless by soluble hydrogen carbonates. Hence if the carbon dioxide liberated by dilute acids from carbonates is allowed to come into contact with phenolphthalein solution colored pink by sodium carbonate solution, it may be identified by the decolourization which takes place.



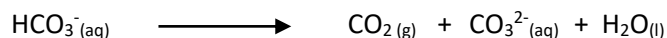
The concentration of the sodium carbonate solution must be such as not to be decolouration under the conditions of the experiment by the carbon dioxide in the atmosphere.

13. Test for hydrogen carbonate (HCO_3^-)

A Freshly prepared 0.5M solution of sodium hydrogen carbonate, NaHCO_3 , or potassium hydrogen carbonate KHCO_3 , can be used to study reactions.

a. Boiling

When boiling hydrogen carbonates decompose



Carbon dioxide, formed can be identified with lime.

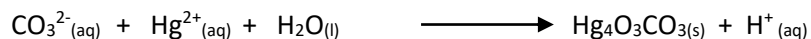
- b. When Magnesium sulphate is added to a cold solution of hydrogen carbonate, no precipitation occurs, while a white precipitate of magnesium carbonate is formed with normal carbonates. On heating the mixture, a white precipitate of Magnesium Carbonate is formed.



Carbon dioxide formed can be identified with lime.

- c. Mercury (II) chloride no precipitate is formed with hydrogen carbonate ions, while in a solution of normal carbonates a reddish – brown precipitate of basic mercury (II) carbonate is formed.

(Caution - mercury (II) chloride is poison)

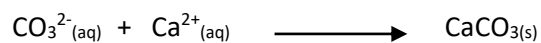


d. Dry test

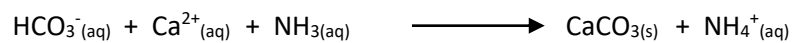
Heating some solid alkali hydrogen carbonate in a dry test tube carbon dioxide is evolved.



- e. Adding an excess of calcium chloride to a mixture of carbonate and hydrogen carbonate the former is precipitated quantitatively.



On filtering the solution rapidly hydrogen carbonate ions pass into the filtrate. On adding ammonia to the filtrate, a white precipitate or cloudiness is obtained if hydrogen carbonates are present.



Reference: Vogel's Qualitative Inorganic Analysis; Seventh edition; page numbers 176-236